ARSLAN MALIK (马林)

arslan@chd.edu.cn | +86 130 2243 4852

@ meetarslanmalik @Arslan Malik

SKILLS

- **Programming:** Python, Java, basic MATLAB
- **AI & LLM Integration:** Prompt engineering, LLM-assisted research and development workflows, Transformer and cross-attention modules, self-supervised representation learning , efficiency, lightweight design, and reliability
- **Deep Learning:** PyTorch (incl. Apple MPS), Ultralytics YOLO, CNNs, model training, debugging and optimization
- **Computer Vision:** Object detection (YOLOv5/v8/v10/v11), small and occluded object detection, multi-object tracking (DeepSORT), person/vehicle Re-ID
- **Efficient AI:** Lightweight network design, real-time inference, feature fusion, CutMix augmentation, adaptive Kalman filtering ,

EDUCATION

- Master's | Chang'an University *CGPA*: *4.0*|5.0 *(July 2025)*
  Field of Study: Computer Science and Technology
  Dissertation title: YOLOv8-SOEP:An improved YOLOv8 Based Model for Detection
- Bachelor | Shenyang University of Chemical Technology *CGPA*: *3.2*|4 *(Jan 2022)*
  Field of Study: Computer Science and Technology

RESEARCH EXPERIENCE

- **Advanced Object Detection Framework using Improved YOLOv8 Architecture**
  - Developed a high-performance detection framework (YOLOV8-SOEP) designed to improve accuracy in complex and cluttered environments.
  - Integrated Spatial Pyramid Dilated Convolution and Omnidirectional Kernel Convolution to strengthen fine-scale feature extraction.
  - Implemented CSP-OmniKernel modules to facilitate multi-directional representation and more efficient feature fusion.
  - Conducted comparative analysis using the VisDrone dataset, achieving superior mAP scores compared to baseline YOLOv8, YOLOv10, and YOLOv11 models.
  - Optimized the architecture for real-time surveillance and autonomous systems with a focus on detecting small or partially occluded objects.

- **DeepSORT-CKFaster: A Lightweight Algorithm for Efficient Trajectory Extraction**
  - Engineered a lightweight deep learning algorithm aimed at maximizing real-time performance and tracking precision.
  - Utilized CutMix data augmentation strategies to address label imbalance and improve model robustness.
  - Introduced an adaptive, feedback-driven feature extraction network to eliminate redundant computations and boost representation quality.
  - Developed an adaptive Kalman filter learning-rate mechanism to stabilize training phases and improve model convergence.
  - Achieved 92.2% average tracking accuracy while reducing total computational costs and training time by over 70%.

ACHIEVEMENTS

- **CSC Scholarship to pursue a Master's** *(2022 – 2025)*
  I was awarded the CSC Scholarship in my master's studies in recognition of my exceptional academic performance and research excellence. This distinction highlights my strong analytical abilities, dedication to high-quality scholarly work, and consistent achievement in a competitive academic setting.
- **Excellent Student Performance Award** *(2024 – 2025)*
  I was honored with the Excellent Student Performance Award, a distinction awarded to top achievers among more than 5,000 students. This recognition reflects my exceptional academic results, strong work ethic, and consistent commitment to excellence throughout my studies.
- **Excellent Graduate Student Award in bachelor's** *(2022)*
  Awarded to the top graduating students based on a comprehensive evaluation of academic excellence (GPA), extracurricular leadership, and research achievements
- **Excellent Student: 2nd Scholarship 10k RMB - 2018** *(2018)*
  Competitive Math, Physics exam

UNIVERSITY SERVICE & COMMUNITY ENGAGEMENT

- **Hainan University Exchange Program 2024 (1-Week Intensive)** *(2024)*
  - Selected for a high-level university exchange focused on cultural immersion and landscape preservation in Hainan.
  - Participated in educational workshops regarding local heritage, contributing to a final report on regional cultural development.
- **Volunteer & Cultural | Joint University Initiative (Xi'an, China)** *(2023)*
  - Collaborated with diverse, international teams to execute community-focused projects, enhancing cross-cultural communication and teamwork.
  - Represented the university during an industrial study tour of Weinan City's Grape Industrial Park, gaining insights into regional agricultural economics and sustainable development practices.
- **Cultural Exchange Participant | Shaanxi Traditional Art Event** *(2023)*
  - Engaged in a collaborative art project with local Shaanxi artists to promote traditional Chinese painting.
  - Facilitated cultural dialogue between local practitioners and international students, deepening institutional ties with the local arts community.

LANGUAGES

- English: Fluent
- Chinese: Intermediate